



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Student Name: DHILIPAN M

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Course Code: CSA0820

Slot: C

Course Name: PYTHON PROGRAMMING FOR SCIENTIFIC COMPUTING

Course Faculty: DR.P.VIJAYARAGAVAN AND DR.ARUNA S K

Project Title: AI-Based Smart Irrigation System Using Python and ESP32

Module Photographs:

MODULE 1

- Developed ESP32 MicroPython firmware to acquire environmental data from connected sensors, uploading readings every 10 seconds to Supabase.
- Implemented a Supabase real-time database to persist sensor data, ensuring sub-100ms latency for dashboard synchronization.
- Designed a Python Flask backend to analyze real-time sensor data, triggering automatic irrigation control via ESP32 when soil moisture drops below 40%.
- Built an interactive React.js web dashboard to visualize sensor trends and pump status, refreshing data every 5 seconds for remote monitoring.
- Integrated the full system, achieving autonomous pump activation from sensor input to mechanical response within 3 seconds for optimized watering.

Dashboard

CSA0820 Python Programming for Scientific Computing

Soil moisture is sufficient

Recommendation: 1

AI-BASED SMART IRRIGATION SYSTEM USING PYTHON AND ESP32

CSA0820 Python Programming for Scientific Computing

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Project Description:

AI-Based Smart Irrigation System Using Python and ESP32 is an innovative IoT-enabled agricultural solution designed to automate irrigation and optimize water usage through real-time environmental monitoring and intelligent decision-making. The system utilizes an ESP32 microcontroller integrated with multiple sensors, including a soil moisture sensor, DHT11 temperature and humidity sensor, rain sensor, water level sensor, ultrasonic sensor, and IR sensor, to continuously monitor field conditions. The ESP32 collects sensor data and transmits it wirelessly to a Python-based application, where the information is processed, stored in a database, and displayed through a web-based dashboard for real-time monitoring. Based on soil moisture levels, weather conditions, and AI-generated recommendations, the system automatically controls a water pump using a relay module, ensuring that crops receive the required amount of water while minimizing water wastage. The system also provides features such as live sensor monitoring, historical data visualization, intelligent irrigation recommendations, water tank level monitoring, and alert notifications for abnormal conditions such as low water levels or animal intrusion. By integrating IoT, Python, database technology, and artificial intelligence, the proposed solution improves irrigation efficiency, reduces manual intervention, conserves water resources, and enhances crop productivity. This smart irrigation system is particularly suitable for agricultural fields, home gardens, greenhouses, research laboratories, and precision farming applications.

Student Signature

Guide Signature